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AI/ML intern

Role: Computer Vision

2D Virtual Try-On (VTON) Model using TPS-Based Fitting

1. Introduction

The objective of this task is to develop a **2D Virtual Try-On (VTON) system** where a clothing image (shirt) is digitally aligned and overlaid onto a person image to simulate the appearance of wearing the garment.

The system focuses on:

- Human pose detection
- Clothing alignment
- Garment fitting
- Background removal
- Image overlay generation

In this implementation, a **2D Virtual Try-On prototype** was developed using **Python, OpenCV, and MediaPipe Pose Detection**. The system estimates body landmarks from the user image and positions the garment according to shoulder and torso measurements.

2. Objective

To create a prototype capable of:

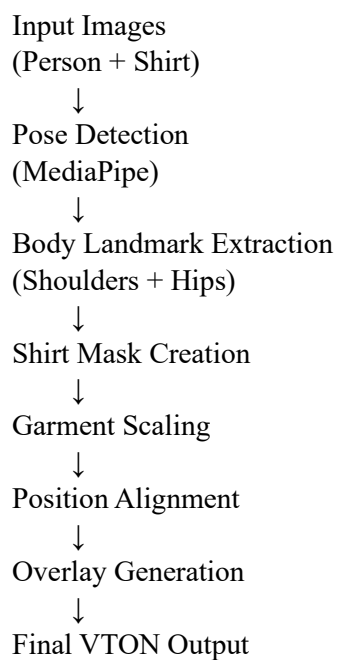
1. Reading person and garment images.
2. Detecting body landmarks.
3. Estimating shoulder and torso dimensions.
4. Aligning shirt image over the body.
5. Generating final virtual try-on output.

3. Technologies Used

Technology Purpose

Python	Development language
OpenCV	Image processing and overlay
MediaPipe	Human pose estimation
NumPy	Matrix operations
Matplotlib	Visualization

4. Methodology / Workflow



5. Code

```
import cv2

import mediapipe as mp

import numpy as np

import matplotlib.pyplot as plt

person = cv2.imread("person.jpg")

shirt = cv2.imread("shirt.png")

if person is None:

    raise Exception("person.jpg missing")
```

```
if shirt is None:
    raise Exception("shirt.png missing")
person_rgb = cv2.cvtColor(
    person,
    cv2.COLOR_BGR2RGB
)
h,w,_ = person.shape
mp_pose = mp.solutions.pose
pose = mp_pose.Pose(
    static_image_mode=True
)
res = pose.process(person_rgb)
if res.pose_landmarks is None:
    raise Exception(
        "Pose not detected"
    )
lm = res.pose_landmarks.landmark
L_SH = lm[
    mp_pose.PoseLandmark.LEFT_SHOULDER
]
R_SH = lm[
    mp_pose.PoseLandmark.RIGHT_SHOULDER
]
L_HIP = lm[
    mp_pose.PoseLandmark.LEFT_HIP
]
R_HIP = lm[
    mp_pose.PoseLandmark.RIGHT_HIP
]
lsx = int(L_SH.x*w)
rsx = int(R_SH.x*w)
sy = int(
```

```

        min(
            L_SH.y,
            R_SH.y
        )*h
    )
    hy = int(
        max(
            L_HIP.y,
            R_HIP.y
        )*h
    )
    gray = cv2.cvtColor(
        shirt,
        cv2.COLOR_BGR2GRAY
    )
    _,mask = cv2.threshold(
        gray,
        180,
        255,
        cv2.THRESH_BINARY_INV
    )
    mask = cv2.GaussianBlur(
        mask,
        (9,9),
        0
    )
    shoulder = abs(
        rsx-lsx
    )
    torso = abs(
        hy-sy
    )

```

```
shirt_w = int(
    shoulder*2.05
)
shirt_h = int(
    torso*1.65
)
shirt = cv2.resize(
    shirt,
    (
        shirt_w,
        shirt_h
    )
)
mask = cv2.resize(
    mask,
    (
        shirt_w,
        shirt_h
    )
)
cx = (
    lsx+rsx
)//2
x1 = cx-shirt_w//2
neck_offset = int(
    shirt_h*0.20
)
y1 = sy-neck_offset
x1 += 5
x2 = x1+shirt_w
y2 = y1+shirt_h
x1=max(0,x1)
```

```
y1=max(0,y1)
x2=min(w,x2)
y2=min(h,y2)
shirt_crop = shirt[
    0:y2-y1,
    0:x2-x1
]
mask_crop = mask[
    0:y2-y1,
    0:x2-x1
]
result = person.copy()
roi = result[
    y1:y2,
    x1:x2
]
mask_f = mask_crop.astype(
    float
)/255.0
mask_f = np.stack(
    [
        mask_f
    ]*3,
    axis=2
)
blend = (
    shirt_crop*mask_f +
    roi*(1-mask_f)
).astype(
    np.uint8
)
result[
```

```
        y1:y2,
        x1:x2
    ] = blend
cv2.imwrite(
    "vton_output.png",
    result
)
plt.figure(
    figsize=(15,6)
)
plt.subplot(
    1,3,1
)
plt.imshow(
    person_rgb
)
plt.title(
    "Boy Image"
)
plt.axis(
    "off"
)
plt.subplot(
    1,3,2
)
plt.imshow(
    cv2.cvtColor(
        shirt,
        cv2.COLOR_BGR2RGB
    )
)
plt.title(
```

```
        "Shirt"
    )
    plt.axis(
        "off"
    )
    plt.subplot(
        1,3,3
    )
    plt.imshow(
        cv2.cvtColor(
            result,
            cv2.COLOR_BGR2RGB
        )
    )
    plt.title(
        "VTON Output"
    )
    plt.axis(
        "off"
    )
    plt.tight_layout()
    plt.show()
```

6. Observations

Observation 1

Pose detection successfully identified:

- Shoulder positions
- Hip positions

using MediaPipe landmarks.

Observation 2

Shirt fitting improves when:

- Neck offset is adjusted
- Shoulder width scaling increases

Observation 3

Masking operation removes background influence and enables smoother overlay.

Observation 4

Alignment accuracy depends on:

- Quality of shirt image
- Transparent background availability

Pose estimation precision

7. Output

Input:

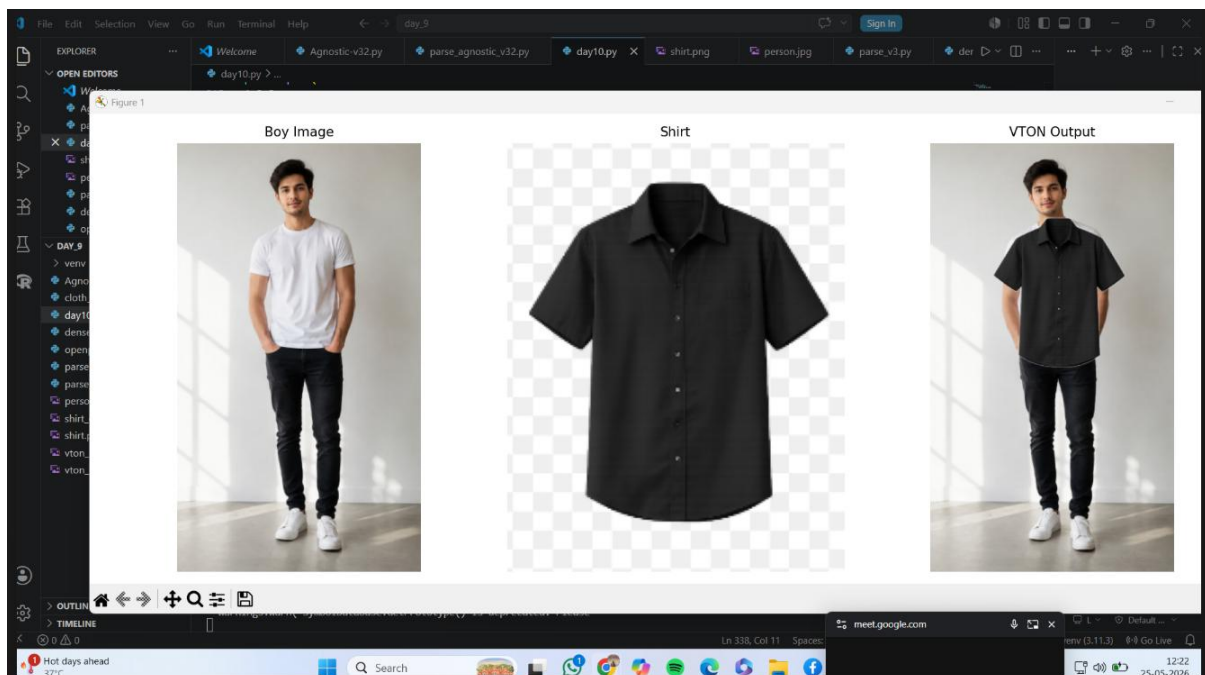
1. Person Image (person.jpg)
2. Shirt Image (shirt.png)

Generated Output:

vton_output.png

Output contains:

- Original person image
- Shirt image
- Final virtual try-on visualization



Result:

The shirt is digitally aligned over the detected torso region to simulate clothing fitting.

8. Advantages

- Simple implementation
- Real-time capable
- Low computational cost
- Easy integration into e-commerce applications
- Useful for fashion recommendation systems

9. Future Improvements

Future enhancements include:

- TPS (Thin Plate Spline) warping
- Human body segmentation
- DensePose integration
- Cloth deformation
- 3D avatar fitting
- Real-time webcam try-on

10. Conclusion

A **2D Virtual Try-On prototype** was successfully implemented using MediaPipe and OpenCV.

The model performs:

- Human pose estimation
- Shirt fitting
- Garment alignment
- Overlay generation

The generated output demonstrates a basic virtual try-on workflow and serves as the foundation for advanced TPS and 3D Virtual Try-On systems.